

1 Introduction

Our notation:

- Random variable X_i with value x_i and possible values from $\mathcal{X}$,
- Random vector $\mathbf{X}$ with range $\mathcal{X}$ and some assignment $\mathbf{x}$,
- Set $\mathcal{A}$, and a measure $\mathbb{M}$.

2 Making Decisions under Uncertainty

3 Convergent Prior

Definition 3.1 (Probabilistic Model). We define a *probabilistic model* $\mathcal{M}$ as a pair $(\mathcal{S}, \mathcal{P})$, where $\mathcal{S}$ is a sample space, and $\mathcal{P}$ is a set of probability measures on $\mathcal{S}$.

Definition 3.2 (Divergence Measure). Let $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \dots\}$ be a set of pairs of discrete random variables in a probabilistic model $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$. We define the *divergence measure* of $P \in \mathcal{P}$ for $\mathcal{X}$ as follows:

$$\mathbb{D}_{\mathcal{X}}[P] = \max_{Q \in \mathcal{P}} \mathbb{E}_Q \left[\sum_i \mathbb{D}(Q_{X_i|e_i} \parallel P_{X_i|e_i}) \right], \quad (1)$$

where $\mathbb{E}_Q[\cdot]$ denotes expectation relative to Q and each $\mathbb{D}(Q_{X_i|e_i} \parallel P_{X_i|e_i})$ is the Kullback-Leibler divergence between $P(X_i|e_i)$ and $Q(X_i|e_i)$:

$$\mathbb{D}(Q_{X_i|e_i} \parallel P_{X_i|e_i}) = \sum_{x_i} Q(x_i | e_i) \ln \frac{Q(x_i | e_i)}{P(x_i | e_i)}.$$

Moreover, we define *the central distribution for $\mathcal{X}$* to be a distribution $C_{\mathcal{X}} \in \mathcal{P}$ which minimizes the divergence measure for $\mathcal{X}$:

$$C_{\mathcal{X}} = \arg \min_{P \in \mathcal{P}} \mathbb{D}_{\mathcal{X}}[P]. \quad (2)$$

Proposition. Using the linearity of expectation, we have:

$$\mathbb{D}_{\mathcal{X}}[P] = \max_{Q \in \mathcal{P}} \sum_i \sum_{x_i, e_i} Q(x_i, e_i) \ln \frac{Q(x_i | e_i)}{P(x_i | e_i)}.$$

Definition 3.3 (Parametric Model). Let Θ be a random vector. A parametric model is a probabilistic model $\mathcal{M}_{\theta} \equiv (\mathcal{S}, \mathcal{P})$, where for every $P \in \mathcal{P}$ there is a prior probability p , such that for every $s \in \mathcal{S}$ we have:

$$P(s) = \int \pi(s | \theta) p(\theta) d\theta, \quad (3)$$

while π is some known probability distribution.

Lemma 1. Let $\mathcal{M}_\theta \equiv (\mathcal{S}, \mathcal{P})$ be a parametric model, and $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \dots\}$ be a set of discrete random variable pairs in $\mathcal{M}_\theta$. For every $P \in \mathcal{P}$ we have:

$$\mathbb{D}_{\mathcal{X}}[P] \leq \max_{\theta \in \Theta} \mathbb{E}_\pi \left[\sum_i \mathbb{D}(\pi_{X_i|e_i, \theta} \parallel P_{X_i|e_i}) \mid \theta \right] \quad (4)$$

$$= \max_{\theta \in \Theta} \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{P(x_i \mid e_i)}. \quad (5)$$

Proof. Let R be an arbitrary distribution in $\mathcal{M}_\theta$, since the Kullback-Leibler divergence is always positive, its expected value is also positive, and for every parameter value θ we have:

$$\sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{R(x_i \mid e_i)} \geq 0.$$

Using this, we can derive the following inequality for every distribution P and every parameter θ :

$$\sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \leq \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{P(x_i \mid e_i)}.$$

Since this inequality holds for any θ , the maximum of the right-hand side is also greater than or equal to the maximum of the left-hand side w.r.t θ :

$$\max_{\theta \in \Theta} \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \leq \max_{\theta \in \Theta} \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{P(x_i \mid e_i)}. \quad (6)$$

$\mathcal{M}_\theta$ is a parametric model and $R(x_i, e_i) = \int \pi(x_i, e_i \mid \theta) r(\theta) d\theta$, while $r(\theta)$ is a probability distribution. Obviously for every θ we have:

$$\sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \leq \max_{\theta \in \Theta} \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)}.$$

Since $r(\theta)$ is a distribution and is non-negative, we can multiply this inequality by $r(\theta)$ and integrate. Notice that the right hand side is not a function of θ and it will remain unchanged because $\int r(\theta) d\theta = 1$:

$$\mathbb{E}_R \left[\sum_i \mathbb{D}(R_{X_i|e_i} \parallel P_{X_i|e_i}) \right] \leq \max_{\theta \in \Theta} \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)}.$$

Now by Inequality 6, for *every* distribution P and R we conclude:

$$\mathbb{E}_R \left[\sum_i \mathbb{D}(R_{X_i|e_i} \parallel P_{X_i|e_i}) \right] \leq \max_{\theta \in \Theta} \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{P(x_i \mid e_i)},$$

which proves the theorem. $\square$

Theorem 2. Let $\mathcal{M}_\theta \equiv (\mathcal{S}, \mathcal{P})$ be a parametric model and $\mathcal{X}$ be a set of pairs of discrete random variables in $\mathcal{M}_\theta$. If for every value of θ there is a distribution $R \in \mathcal{P}$ such that $R(s) = \pi(s \mid \theta)$ for every $s \in \mathcal{S}$, then for any $P \in \mathcal{P}$ we have:

$$\mathbb{D}_{\mathcal{X}}[P] = \max_{\theta \in \Theta} \mathbb{E}_\pi \left[\sum_i \mathbb{D}(\pi_{X_i|e_i, \theta} \parallel P_{X_i|e_i}) \mid \theta \right]. \quad (7)$$

Proof. For an arbitrary distribution $P \in \mathcal{P}$, we let:

$$\theta^* = \arg \max_{\theta \in \Theta} \mathbb{E}_\pi \left[\sum_i \mathbb{D}(\pi_{X_i|e_i, \theta} \parallel P_{X_i|e_i}) \mid \theta \right].$$

In $\mathcal{M}_\theta$ there should be a distribution $R \in \mathcal{P}$ that equals $\pi(\cdot \mid \theta^*)$, and we have:

$$\mathbb{E}_R \left[\sum_i \mathbb{D}(R_{X_i|e_i} \parallel P_{X_i|e_i}) \right] = \mathbb{E}_\pi \left[\sum_i \mathbb{D}(\pi_{X_i|e_i, \theta^*} \parallel P_{X_i|e_i}) \mid \theta^* \right].$$

On the other hand, we see:

$$\max_{Q \in \mathcal{P}} \mathbb{E}_Q \left[\sum_i \mathbb{D}(Q_{X_i|e_i} \parallel P_{X_i|e_i}) \right] \geq \mathbb{E}_R \left[\sum_i \mathbb{D}(R_{X_i|e_i} \parallel P_{X_i|e_i}) \right],$$

Thus:

$$\mathbb{D}_{\mathcal{X}}[P] \geq \max_{\theta \in \Theta} \mathbb{E}_\pi \left[\sum_i \mathbb{D}(\pi_{X_i|e_i, \theta} \parallel P_{X_i|e_i}) \mid \theta \right].$$

This and Inequality 5 proves the theorem. $\square$

Theorem 3 (Minimax theorem [1]). Let $\mathcal{X} \subset \mathbb{R}^n$ and $\mathcal{Y} \subset \mathbb{R}^m$ be compact convex sets. If $f : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ is a continuous function that is concave-convex, i.e.

- $f(\cdot, y) : \mathcal{X} \rightarrow \mathbb{R}$ is concave for fixed y , and
- $f(x, \cdot) : \mathcal{Y} \rightarrow \mathbb{R}$ is convex for fixed x .

Then we have that:

$$\max_{x \in \mathcal{X}} \min_{y \in \mathcal{Y}} f(x, y) = \min_{y \in \mathcal{Y}} \max_{x \in \mathcal{X}} f(x, y). \quad (8)$$

Theorem 4. Let $\mathcal{M}_\theta \equiv (\mathcal{S}, \mathcal{P})$ be a parametric model which satisfies the conditions of Theorem 2, and $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \dots\}$ be a set of pairs of discrete random variables in $\mathcal{M}_\theta$. The central distribution for $\mathcal{X}$ in $\mathcal{M}_\theta$ can be obtained by:

$$C_{\mathcal{X}} = \max_{P \in \mathcal{P}} \int \mathbb{E}_\pi \left[\sum_i \mathbb{D}(\pi_{X_i|e_i, \theta} \parallel P_{X_i|e_i}) \mid \theta \right] p(\theta) d\theta \quad (9)$$

where $p(\theta)$ is the prior distribution for P .

Proof. $\square$

References

- [1] Ding-Zhu Du. *Minimax and Its Applications*, pages 339–367. Springer US, Boston, MA, 1995.